POWER BI

SALES & PROFIT ANALYSIS

INTRODUCTION

This practical assignment demonstrates the use of Power BI to clean up a sales dataset, create meaningful visualizations, and apply DAX calculations. The objective is to transform the raw sales data into a clear report that can be used to understand order distribution, product performance, and profit trends.

Objectives

- Clean and prepare the sales dataset for analysis.
- Create visualizations to understand regional orders, product trends, and profit over time.
- Create a calculated table, calculated column, and measured using DAX.
- Present the analysis in an interactive Power BI report.

Dataset

The dataset used for the practical is SalesData_1000Rows_WithIssues_copy.csv. The dataset contains order information including OrderID, OrderDate, CustomerName, Region, Product, Category, Quantity, UnitPrice, Sales, Cost, and Profit.

Q1. DATA CLEANING & LOADING

1. Importing the Dataset

The CSV file was imported into Power BI Desktop using Get Data → Text/CSV. The data was opened in Power Query using Transform Data so that the required cleaning operations could be performed before loading.

2. Standardizing OrderDate

The OrderDate column was checked and its data type was changed to Date. Rows with missing OrderDate values were filtered out so that the date field could be used correctly in the time-based visualization.

3. Removing Duplicate Records

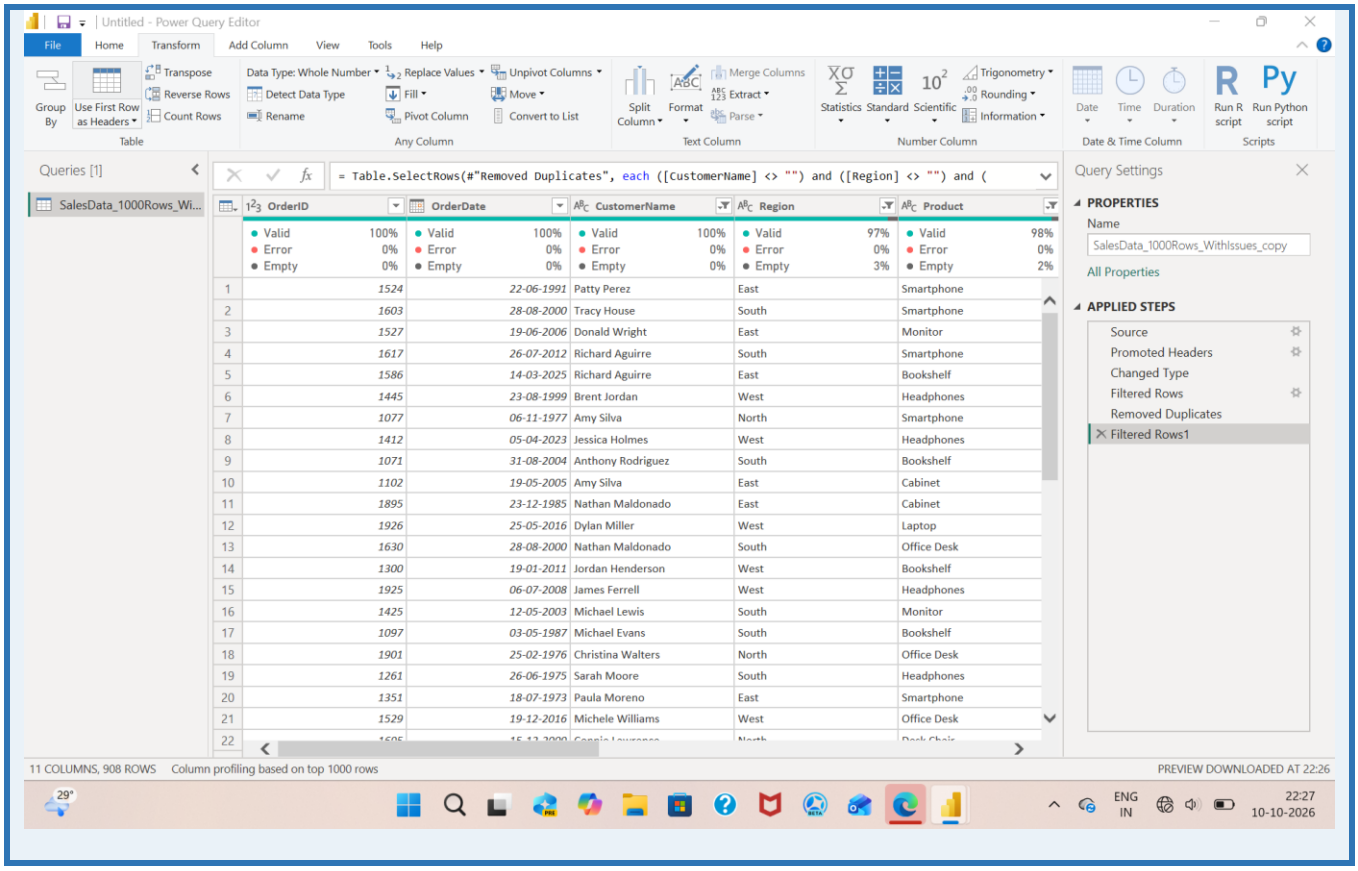
Duplicate rows were removed using the Remove Duplicates option in Power Query. This prevents repeated records from affecting order counts and other calculations.

4. Handling Missing Values

Rows containing missing values in the important analysis fields were removed. The cleaned dataset was checked for missing values in CustomerName, Region, Product, UnitPrice, Sales, Cost, and Profit.

5. Loading the Cleaned Data

After completing the cleaning steps, Close & Apply was selected to load the cleaned data into Power BI Desktop. The cleaned table was then used for the visualizations and DAX calculations.


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [1]
SalesData_1000Rows_Wi...
= Table.SelectRows(#"Removed Duplicates", each ([CustomerName] <> "") and ([Region] <> "") and (
OrderID
OrderDate
CustomerName
Region
Product
Query Settings
PROPERTIES
Name
SalesData_1000Rows_WithIssues_copy
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Filtered Rows
Removed Duplicates
Filtered Rows1
11 COLUMNS, 908 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 22:26
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10-10-2026

Q2. VISUALIZATIONS

Three visuals were created in Power BI to analyze the cleaned sales data. The visuals were arranged together as a dashboard so that the main sales and profit information could be viewed easily.

2.1 Order Distribution by Region

A Pie Chart was created to show how orders are distributed across different regions. Region was placed in the Legend field and Count of OrderID was used as the Values field. Data labels were enabled to make the distribution easier to read.

Purpose

This visual helps compare the number of orders received from each region and quickly identifies which regions contribute more or fewer orders.

2.2 Top 5 Products by Orders

A Clustered Column Chart was created to identify the five products with the highest number of orders. Product was used on the X-axis and Count of OrderID was used on the Y-axis. A Top 5 filter was applied to display only the five leading products.

Purpose

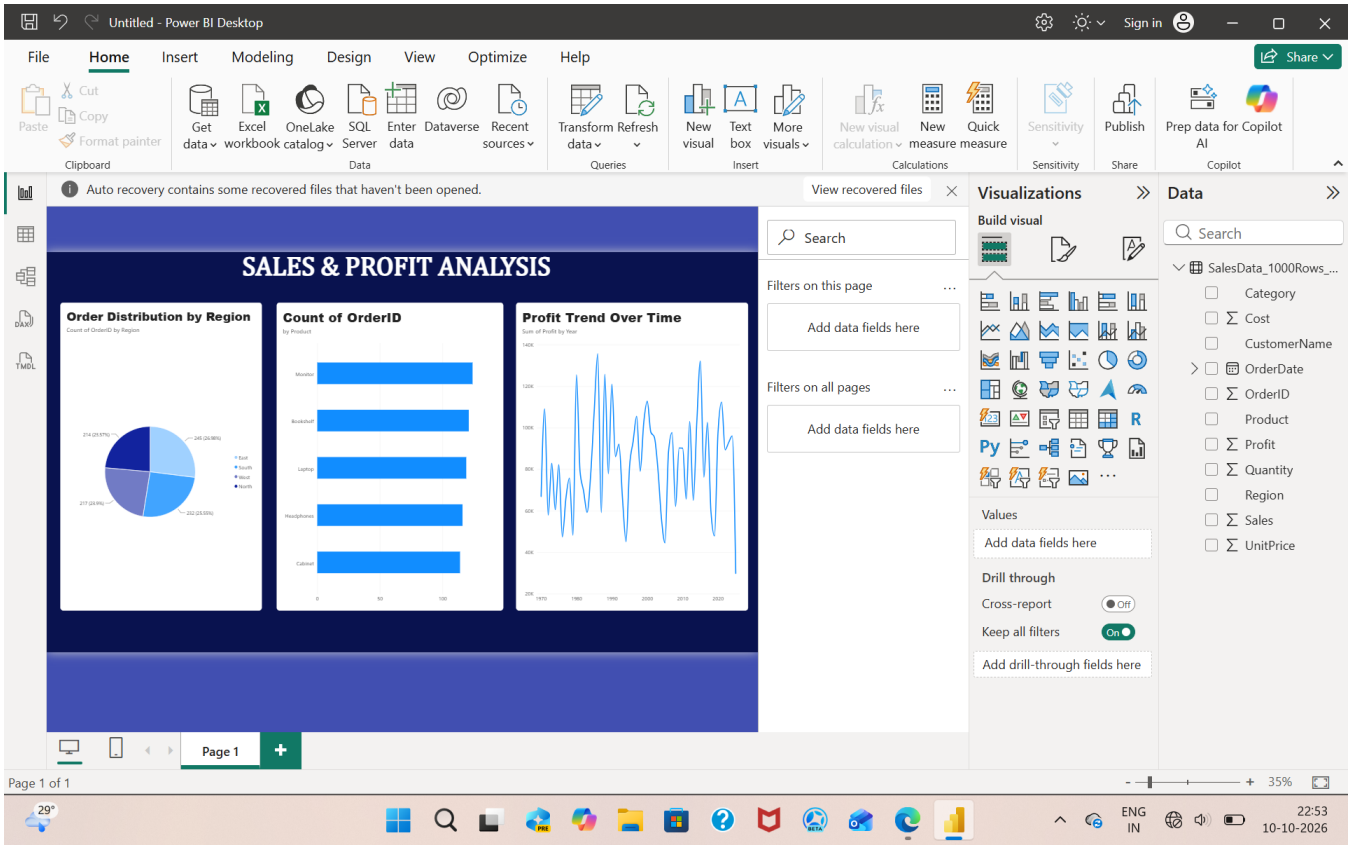
This visual highlights the most frequently ordered products and provides a simple way to compare their order volumes.

2.3 Profit Trend Over Time

A Line Chart was created to show how profit changes over time. OrderDate was used on the X-axis and Sum of Profit was used on the Y-axis. The date was used as a plain date field so the trend could be displayed across time.

Purpose

This visual helps observe changes in profit over time and makes increases or decreases in profitability easier to identify.


Untitled - Power BI Desktop
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Sensitivity
Publish
Share
Prep data for AI
Copilot
Auto recovery contains some recovered files that haven't been opened.
View recovered files
SALES & PROFIT ANALYSIS
Order Distribution by Region
Count of OrderID
Profit Trend Over Time
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Data
Search
SalesData_1000Rows_...
Category
Cost
CustomerName
OrderDate
OrderID
Product
Profit
Quantity
Region
Sales
UnitPrice
Page 1
Page 1 of 1
35%
29°
ENG
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22:53
10-10-2026

Q3. DAX CALCULATIONS

DAX was used to create a calculated table, a calculated column, and a measure. These calculations extend the analysis beyond the basic visualizations.

3.1 Calculated Table – EastRegionOrders

The EastRegionOrders table was created by filtering the main sales table so that only records where Region equals East are included.

```
EastRegionOrders =
FILTER(
    'SalesData_1000Rows_WithIssues_Copy',
    'SalesData_1000Rows_WithIssues_Copy'[Region] = "East"
)
```

3.2 Calculated Column – ProfitCostDifference

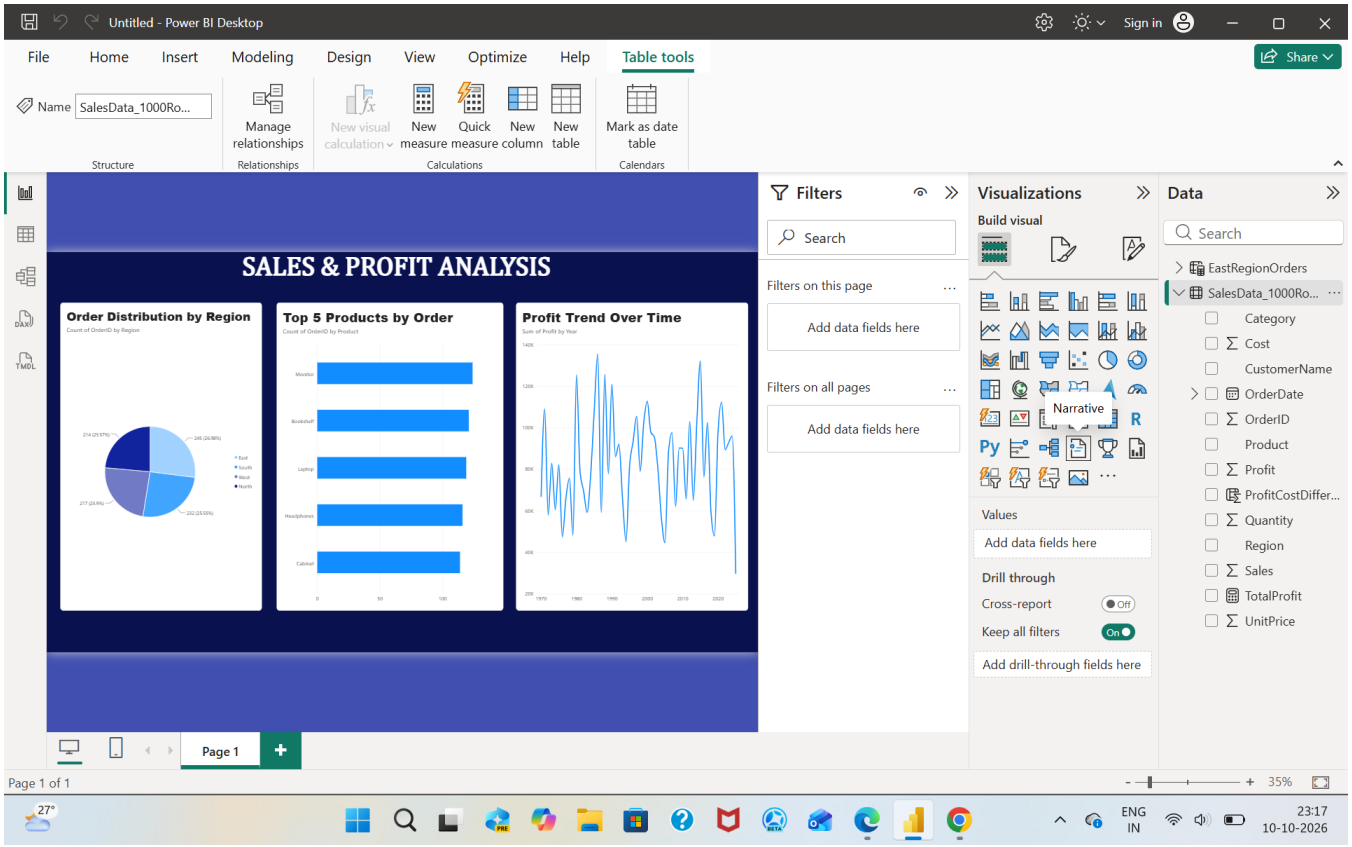
The ProfitCostDifference calculated column was created by subtracting Cost from Profit for each row. This provides a row-level comparison between profit and cost.

```
ProfitCostDifference =
SalesData_1000Rows_WithIssues_Copy[Profit] -
SalesData_1000Rows_WithIssues_Copy[Cost]
```

3.3 Measure – TotalProfit

The TotalProfit measure was created using SUM on the Profit column. It calculates the total profit for the current report context.

```
TotalProfit =
SUM(SalesData_1000Rows_WithIssues_Copy[Profit])
```


Untitled - Power BI Desktop
Sign in
File
Home
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Modeling
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View
Optimize
Help
Table tools
Share
Name
SalesData_1000Ro...
Manage relationships
New visual calculation
New measure
Quick measure
New column
New table
Mark as date table
Structure
Relationships
Calculations
Calendars
SALES & PROFIT ANALYSIS
Order Distribution by Region
Top 5 Products by Order
Profit Trend Over Time
Filters
Search
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Narrative
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Data
Search
EastRegionOrders
SalesData_1000Ro...
Category
Cost
CustomerName
OrderDate
OrderID
Product
Profit
ProfitCostDiffer...
Quantity
Region
Sales
TotalProfit
UnitPrice
Page 1
Page 1 of 1
35%
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10-10-2026

RESULTS & OBSERVATIONS

Data Preparation

The raw dataset was cleaned by standardizing the date field, removing duplicates, and removing records with missing values in the required fields. This produced a cleaner dataset for reporting.

Visual Analysis

- The regional pie chart provides a quick comparison of order distribution.
- The Top 5 Products chart identifies the products with the highest order frequency.
- The profit line chart shows the movement of profit over time.

DAX Analysis

- The EastRegionOrders calculated table isolates records belonging to the East region.
- The ProfitCostDifference calculated column provides a row-level profit and cost comparison.
- The TotalProfit measure calculates total profit dynamically in the report.

CONCLUSION

The Power BI practical was completed by importing and cleaning the sales dataset, creating three required visualizations, and implementing the required DAX calculations. The final dashboard presents regional order distribution, the top five products by orders, and profit trends over time. The DAX calculations further support focused regional analysis and profit evaluation.